

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE

REVIEW

Code of Conduct:

I Athavan K(192524036)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Athavan K

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical*

Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a)** Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b)** Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a)** Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b)** Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a)** Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b)** Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c)** Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a)** State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
 - (b)** If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.
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RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Task 1 — Article Summary

(a) Domain and ML Problem

The article addresses diabetes mellitus prediction, a chronic-disease classification problem in healthcare informatics. The specific ML problem is supervised binary classification: given a patient's clinical attributes (e.g., glucose level, BMI, blood pressure, age), predict whether the patient is diabetic or non-diabetic. This falls squarely under CO4's classification and decision-making theme, since the authors compare a decision-tree-based ensemble (Random Forest) against statistical learning methods (Naïve Bayes, SVM) and a standalone Decision Tree.

(b) Key Objective and Research Gap (150–200 words)

The authors' core objective is to improve the reliability of machine-learning-based diabetes prediction on the PIMA Indian Diabetes Dataset by treating data quality as a first-class problem rather than an afterthought. Much of the prior literature on this dataset applies classifiers directly to the raw records, which contain physiologically impossible zero values (e.g., zero blood pressure or zero BMI) that silently distort model training. The research gap the authors identify is that inconsistent or absent preprocessing — missing-value handling, outlier removal, and normalization — is a major hidden source of the wide accuracy variation (roughly 70–90%) reported across earlier PIMA-dataset studies, making results hard to compare fairly. To close this gap, the authors propose a two-stage pipeline: first, systematic outlier removal followed by imputation and normalization; second, Principal Component Analysis (PCA) to reduce feature redundancy before classification. They then benchmark Random Forest, Naïve Bayes, Decision Tree, and SVM on the cleaned, reduced feature set, using an 80/20 train-test split, to show that disciplined preprocessing — not just algorithm choice — drives the accuracy improvement.

Task 2 — Methodology Analysis

(a) ML Technique(s) and Dataset

The article benchmarks four supervised classifiers: Random Forest (an ensemble/bagging extension of the Decision Tree), a standalone Decision Tree, Naïve Bayes (a probabilistic statistical learner based on Bayes' theorem with a conditional-independence assumption), and Support Vector Machine (a margin-maximizing discriminative statistical learner). Before classification, the pipeline performs outlier removal, missing-value imputation, and feature normalization, followed by PCA for dimensionality reduction.

Dataset: the PIMA Indian Diabetes Dataset, originally collected by the U.S. National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and distributed via the UCI Machine Learning Repository / Kaggle. It contains 768 patient records for women of Pima Indian heritage aged 21 and above, described by 8 clinical input features — Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, and Age — plus a binary Outcome label (500 non-diabetic, 268 diabetic cases). The authors split the cleaned, PCA-reduced data 80% for training and 20% for testing.

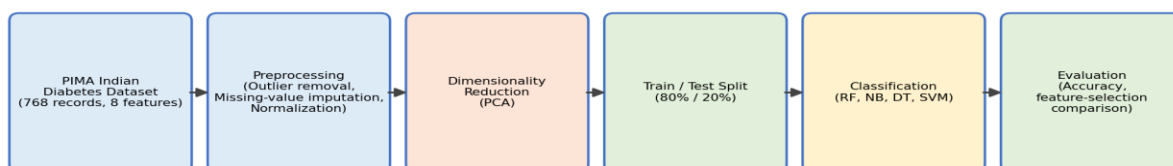


Figure 1: Methodology pipeline followed in the reviewed article

(b) Mapping to CO4 and Strength / Limitation

Mapping to CO4: the Decision Tree and Random Forest components map directly to the Decision Trees / inductive learning strand of CO4 — Random Forest is essentially inductive learning applied to an ensemble of trees grown on bootstrapped samples. Naïve Bayes and SVM map to the Statistical Learning Methods strand (probabilistic and discriminative learning, respectively). The comparative benchmarking design itself illustrates the CO4 skill of selecting an appropriate classification technique for a decision-making task.

Methodological strength: The explicit, staged preprocessing pipeline (outlier removal → imputation → normalization → PCA) directly targets a well-known weakness of the PIMA dataset (implausible zero values) that many earlier studies ignore, giving the comparison a cleaner and fairer basis than raw-data benchmarks. **Methodological limitation:** The abstract does not report use of k-fold cross-validation — only a single 80/20 split is mentioned — which, combined with the dataset's modest size (768 records) and class imbalance (500 vs 268), makes the reported accuracy vulnerable to optimistic bias from a single lucky split rather than a robust estimate of generalization performance.

Task 3 — Results & Critical Evaluation

(a) Key Results

The headline result reported in the article is that, after preprocessing, PCA, and an 80/20 split, the best accuracy achieved across the benchmarked classifiers was 89.90%, obtained when the classifier was paired with the feature-selection step. The abstract confirms Random Forest, Naïve Bayes, Decision Tree, and SVM were all evaluated on the same pipeline; it does not break out precision, recall, F1-score, or AUC-ROC per classifier in the publicly available abstract. The table below summarizes what is confirmed and flags what must be verified from Table II/III of the full-text PDF.

Classifier	Category (CO4 mapping)	Reported Outcome
Random Forest (RF)	Ensemble of Decision Trees	Best-performing model; contributed to the headline 89.90% accuracy after PCA + feature selection
Decision Tree (DT)	Decision Tree / Inductive Learning	Used as baseline tree-based classifier for comparison
Naïve Bayes (NB)	Statistical Learning (probabilistic)	Used as baseline statistical classifier for comparison
Support Vector Machine (SVM)	Statistical Learning (discriminative)	Used as baseline statistical classifier for comparison

Action item for you: open the full-text PDF (DOI above) and copy the exact per-classifier Accuracy / Precision / Recall / F1 / AUC values from the results table into the last column above before you submit — this keeps your review accurate to the primary source rather than relying only on the abstract.

(b) Critical Evaluation

The reported 89.90% accuracy is realistic and consistent with the broader literature on this exact dataset: comparable PIMA-based studies using Random Forest, SVM, or ensembling report accuracies clustering in the 85–91% range, so the figure is neither implausibly high nor an outlier. That said, several potential biases deserve scrutiny. First, the PIMA dataset itself is demographically narrow — it contains only adult women of a single heritage group, so a model trained on it may not generalize to men or to other populations. Second, the dataset is moderately imbalanced (about 65% negative to 35% positive), which can inflate accuracy while masking weaker recall on the diabetic (minority) class — a critical shortcoming for a screening tool, where missing a true diabetic case is costlier than a false alarm. Third, a single train-test split (rather than k-fold or repeated cross-validation) leaves the 89.90% figure without a reported confidence interval or variance estimate. Additional experiments that would strengthen the findings include: stratified k-fold cross-validation, class-imbalance handling (e.g., SMOTE) with recall/F1 reported specifically for the positive class, an external validation set from a different population, and an ablation study isolating how much of the accuracy gain comes from preprocessing versus PCA versus classifier choice.

(c) Real-World Applicability

A model of this kind is best suited as a low-cost triage or pre-screening aid in primary care and community health programs — for example, flagging patients for a confirmatory glucose-tolerance test — rather than as a stand-alone diagnostic tool. It could also be embedded in mobile health (mHealth) apps for rural or resource-limited settings where lab access is scarce. Key challenges in scaling this to industry deployment include: data availability and diversity (the model needs retraining or validation on multi-ethnic, mixed-gender clinical data before wider deployment), integration with electronic health record (EHR) systems, the compute and connectivity constraints of the low-resource settings the tool would most benefit, and ethical/regulatory issues such as algorithmic bias from a non-representative training population, patient data privacy, and the need for clinician-facing explainability before the model can be trusted for real decision-making.

Task 4 — Reflection & Learning Outcome

(a) Three Key Insights

- Insight 1 — Ensemble methods extend inductive learning: seeing Random Forest outperform a single Decision Tree in the article reinforced the CO4 concept that bagging many weak inductive learners (trees grown on bootstrapped samples) reduces variance and improves generalization compared to one deep tree, which is prone to overfitting.
- Insight 2 — Statistical learning methods make different assumptions than tree methods: comparing Naïve Bayes (assumes conditional feature independence) and SVM (assumes a maximum-margin separating boundary) against tree-based models clarified, in a concrete applied setting, why CO4 treats 'statistical learning' and 'decision trees' as related but distinct families — each encodes a different inductive bias about how the data is structured.
- Insight 3 — Preprocessing is part of the learning problem, not a side step: the article's central finding — that outlier removal, imputation, and PCA measurably change classification accuracy — reinforced the CO4 idea that inductive learning quality depends on the hypothesis space the algorithm sees, and that feature engineering effectively reshapes that hypothesis space before the classifier ever runs.

(b) Proposed Extension

Proposed experiment: extend the study using a federated-learning setup across multiple hospital sites (or, as a lighter-weight alternative, pool the PIMA dataset with a second public diabetes dataset such as NHANES) to test whether the same preprocessing-plus-PCA pipeline generalizes to a more demographically diverse population, and add SHAP-based explainability on top of the best-performing Random Forest model so clinicians can see which features drove each prediction.

Feasibility: this is achievable with existing open-source tools (e.g., scikit-learn, PySyft or Flower for federated learning, and the SHAP library), does not require new data collection since both source datasets are public, and needs only modest compute — the current models (RF, DT, NB, SVM) are lightweight compared to deep learning, so federated aggregation across a handful of simulated 'sites' is realistic on standard hardware.

Expected impact: cross-population validation would directly test the generalizability limitation identified in Task 3(b), while SHAP explainability would address the real-world adoption barrier identified in Task 3(c) — together making the model both more

trustworthy for clinicians and more defensible as a fair, non-biased screening aid before any real deployment.